Fiscal Year:	FY 2015	**Task Last Updated:**	FY 04/27/2016
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	08/01/2013	**End Date:**	07/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2016 per NSSC information (Ed., 7/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AK69G		
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Performance Goal Text:			
	[Editor's note 11/11/2013. Note this is a continuation of "High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers" (grant NNX12AE79G); see that project for previous reports.] This is a collaborative project with other NASA investigators who are characterizing mouse models of human lung and colon cancers. Data from both our laboratory and the literature suggest that epigenetics play a significant role in carcinogenic processes. By describing the changes in DNA methylation and miR regulation of gene expression for these mice, we will provide NASA with mechanistic data regarding carcinogenic initiation and progression. By using normal, cancer, and metastatic cancer cell lines we will obtain information regarding epigenetic changes that occur during carcinogenesis based on stage. For example, loss of control of gene expression for extracellular matrix (ECM) remodeling genes may foster metastatic behavior in colon cancer. For this reason we will include ECM gene promoter		

Task Description:	DNA methylation and gene expression among the loci evaluated in this study to the hypothesis that dis-regulation of epigenetic marks and gene expression changes with disease progression. Similarly, the work of other NASA investigators suggests that we will see epigenetically regulated differences among cell lines in expression of inflammatory response pathway gene regulation among lung cell lines. These experiments will provide information regarding radiation responses of these mouse tissues and human cell lines based on the LET (linear energy transfer) of the radiation used. Abnormal epigenetic silencing of genes can occur at a variety of times during tumor progression. If so, does radiation exposure push normal cells to a cancer phenotype and cancer cells to a metastatic phenotype? This question is of importance to NASA since astronauts are of an age where they likely go into missions with precursor lesions. It is important to understand whether their occupational radiation exposures lead to increased cancer risk. Tasks: 1. Evaluate normal and cancer cell lines from lung and colon for responses to low dose high LET radiation exposure 2. Establish epigenetic profiles for in vivo radiation induced cancers in mouse models of human lung and colon cancers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reducing the uncertainties associated with cancer risk models for radiation exposures involved in space flight is of considerable importance in light of long-term occupation of the International Space Station (ISS) and other future manned missions by astronauts. A greater understanding of these carcinogenic mechanisms also benefits Earth-bound populations. Colon cancer is the third most common cancer. Precancerous lesions in the GI tract are found in approximately 1 out of every 10 people by age 40 and this incidence increases significantly with age. In addition to being the largest target for radiation effects, lung cancer is the most prevalent form of cancer in men and women. Each of these cancers has hallmark mutations associated with them. On top of these genetic changes, our understanding of carcinogenesis is complicated by epigenetic alterations to the genome. These epigenetic changes result in misregulation of gene expression for pathways such as cell cycle regulation, DNA repair, apoptosis, and extracellular matrix remodeling. A better understanding of these epigenetic mechanisms may lead to improvements in clinical treatments for lung, colon, and other cancers.
Task Progress:	As mentioned in last year's report, my laboratory has begun evaluating a novel mechanism that has been implicated in carcinogenic processes. Exosomes are cell secreted microvesicles that are highly enriched in biological effectors of carcinogenesis such as growth factors, cytokines, mRNA, miR, and bioreactive lipids. Recently it has been shown that exosomes play an essential role in the growth and/or metastasis of numerous cancers of the colon, prostate, lung, breast, and brain. Exosomal signaling has been shown to alter the cellular phenotype of target cells and a positive correlation between the abundance of secreted exosomes and cancer progression has been demonstrated. The release of exosomes is also stimulated by exposure to ionizing radiation. While not a component of the original project, we believe that exosome secretion is going to prove to play an important role in the radiation response and cancer processes. For this reason we have initiated experiments to test the hypothesis that exosomes secreted by irradiated cells contain unique miR and mRNA cargo that can alter the phenotype of target cells.
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Articles in Peer-reviewed Journals	Lasio G, Guerrero M, Goetz W, Lima F, Baulch JE. "Effect of varying dose-per-pulse and average dose rate in X-ray beam irradiation on cultured cell survival." Radiat Environ Biophys. 2014 Nov;53(4):671-6. http://dx.doi.org/10.1007/s00411-014-0565-2 ; PubMed PMID: 25169705 , Nov-2014
Articles in Peer-reviewed Journals	Baulch JE, Aypar U, Waters KM, Yang AJ, Morgan WF. "Genetic and epigenetic changes in chromosomally stable and unstable progeny of irradiated cells." PLoS One. 2014 Sep 24;9(9):e107722. eCollection 2014. http://dx.doi.org/10.1371/journal.pone.0107722 ; PubMed PMID: 25251398; PubMed Central PMCID: PMC4175465 , Sep-2014
Articles in Peer-reviewed Journals	Keszenman DJ, Kolodiuk L, Baulch JE. "DNA damage in cells exhibiting radiation-induced genomic instability." Mutagenesis. 2015 May;30(3):451-8. Epub 2015 Feb 22. http://dx.doi.org/10.1093/mutage/gev006 ; PubMed PMID: 25711497 , May-2015
Articles in Peer-reviewed Journals	Baulch JE, Craver BM, Tran KK, Yu L, Chmielewski N, Allen BD, Limoli CL. "Persistent oxidative stress in human neural stem cells exposed to low fluences of charged particles." Redox Biol. 2015 Aug;5:24-32. Epub 2015 Mar 11. http://dx.doi.org/10.1016/j.redox.2015.03.001 ; PubMed PMID: 25800120; PubMed Central PMCID: PMC4371546 , Aug-2015